

Ruthvik Reddy Veerannagari

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PROFESSIONAL SUMMARY

Computer Science (Data Science) graduate with research experience in Artificial Intelligence, Machine Learning, Computer Vision, Graph Neural Networks, and Extended Reality. Experienced in developing scalable backend systems using Django, FastAPI, PostgreSQL, and blockchain technologies while contributing to research in intelligent wireless communications and immersive AI applications. Author of research manuscripts submitted to international journals and conferences.

TECHNICAL SKILLS

Languages	Python, C/C++, SQL
Backend	Django, FastAPI, REST API Development, Authentication (JWT/OTP), Session Management
Frontend	Next.js, HTML5, CSS3, Bootstrap
AI / ML	Machine Learning, Deep Learning, Graph Neural Networks (GNN), Reinforcement Learning (DRL), NLP, Computer Vision, Semantic Scene Generation, Feature Engineering, Model Evaluation
Blockchain	Ethereum, Solidity, Hardhat, Ethers.js, IPFS, Smart Contracts, SHA-256 Cryptographic Hashing
XR / Visualization	Extended Reality (XR), Augmented Reality (AR), 3D Scene Graph Generation, Scientific Knowledge Visualization, Adaptive Learning Systems
Databases	PostgreSQL, Relational Database Design, Schema Design, Query Optimization
Libraries	TensorFlow, OpenCV, Pandas, NumPy, Matplotlib, Scikit-learn
Tools	Git, GitHub, Linux, Jupyter Notebook, Google Colab, VS Code, Postman

PROFESSIONAL EXPERIENCE

Software Developer Intern — Cypwng, LendFundz Fintech Platform, Hyderabad

Jan 2025 – Jun 2025

- Engineered backend modules for loan origination, approval workflows, repayment tracking, and status management using Django and PostgreSQL, improving operational efficiency across the lending platform.
- Implemented secure authentication using OTP-based two-factor verification, role-based access control, and session management to satisfy production security and compliance requirements.
- Designed and developed RESTful APIs for seamless integration with a Next.js frontend, enabling parallel development and scalable client-server communication.
- Built modular EMI calculation and repayment scheduling services supporting multiple loan products while improving maintainability, numerical accuracy, and regression reliability.

Research Intern — IIIT Naya Raipur

Jan 2026 – Present

- Conducting research on AI-driven wireless communication systems, intelligent networking, and multimodal machine learning as part of the NMICPS-funded project on AI/ML-based Network Management and Signal Processing for 5G and Beyond.
- Developed end-to-end machine learning pipelines involving data preprocessing, feature engineering, model training, hyperparameter optimization, and reproducible experimental evaluation.
- Implemented and evaluated deep learning, graph neural network, and reinforcement learning models for intelligent resource allocation, signal processing, and wireless communication applications.
- Contributed to scientific publications through literature review, experimental design, implementation, quantitative performance analysis, and preparation of research manuscripts for international journals and conferences.

PUBLICATIONS (DRAFTED / UNDER REVIEW)

[1] Ruthvik Reddy Veerannagari, "EduChain: A Blockchain-Based Framework for Decentralised Academic Credential Verification," *Computer Networks*, Under Review, 2026.

[2] Ruthvik Reddy Veerannagari, "Scalable Resource Allocation Using Graph Learning in 6G for Hyper-Reliable V2X Communication," *IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, Under Review, 2026.

[3] Ruthvik Reddy Veerannagari, "ArchViz-XR: AI-Powered Extended Reality Framework for Scientific Knowledge Visualization," Draft Manuscript, 2026.

TECHNICAL PROJECTS

ArchViz-XR | AI-Powered XR Framework for Scientific Knowledge Visualization

- Designed an automated *semantic-to-spatial XR compilation pipeline* that transforms research paper diagrams into immersive 3D AR experiences using NLP, computer vision, and scene graph generation.
- Developed multimodal AI modules for entity extraction, relationship reasoning, spatial scene construction, and adaptive content generation, eliminating manual AR authoring.
- Implemented interactive visualization with AI-generated explanations, voice interaction, and real-time exploration of Transformer architectures and scientific workflows.

GNN-DRL & GNN-SPN | AI-Based Resource Allocation for V2X Networks

- Implemented Graph Neural Network integrated Deep Reinforcement Learning (GNN-DRL) and Graph Neural Network with Successive Policy Networks (GNN-SPN) for dynamic radio resource allocation in high-mobility V2X communication networks.
- Evaluated performance under varying vehicle densities using V2I throughput, latency, interference management, packet delivery ratio, and reliability metrics.
- Conducted comparative analysis against conventional DRL approaches, demonstrating improved spectrum utilization, network stability, and communication efficiency through graph-based learning.

EduChain | Decentralised Academic Credential Verification Platform

- Developed a blockchain-based credential verification platform using Ethereum smart contracts for secure certificate issuance, verification, and revocation.

- Integrated IPFS with SHA-256 cryptographic hashing to provide tamper-evident storage while enabling decentralized document retrieval.
- Built a FastAPI backend with a wallet-authenticated Next.js frontend using Ethers.js, enabling trustless verification without centralized intermediaries.

LendFundz | Financial Management System

- Developed a fintech platform supporting end-to-end loan lifecycle management, approval workflows, EMI computation, and repayment tracking.
- Designed normalized PostgreSQL schemas, secure Django REST APIs, OTP-based authentication, and role-based access control for financial applications.
- Optimized backend services for scalability, modularity, and seamless integration with the Next.js frontend.

ParkEase | Real-Time Smart Parking Allocation System

- Developed a computer vision-based smart parking system using OpenCV to detect parking slot occupancy from CCTV feeds and provide real-time availability updates.
- Implemented automated parking allocation, reservation management, and concurrency control mechanisms to prevent duplicate bookings and maintain data consistency.
- Designed scalable backend logic for live parking status monitoring, improving parking utilization and reducing vehicle search time in smart-city environments.

Mental Health Detector | Journal Sentiment Analysis Application

- Built a supervised NLP pipeline for sentiment classification of personal journal entries to monitor emotional and behavioral patterns.
- Applied text preprocessing, feature engineering, and machine learning techniques to classify user sentiment with interpretable predictions.
- Developed Matplotlib-based dashboards for longitudinal mood visualization and behavioral trend analysis.

EDUCATION

B.Tech (Honours) in Computer Science Engineering (Data Science)

2022 – 2026

Chhattisgarh Swami Vivekanand Technical University, Bhilai

RESEARCH INTERESTS

Seeking PhD opportunities in Artificial Intelligence and Wireless Communications with research interests in:

- 5G/6G Wireless Communication Systems
- Machine Learning and Artificial Intelligence for Wireless Networks
- Internet of Things (IoT) and Intelligent Edge Computing
- Physical Layer Security and Trustworthy Communication Systems
- Wireless Resource Allocation using Graph Neural Networks and Reinforcement Learning

CERTIFICATIONS

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|---|---|
| • Machine Learning with Python — IBM | • Advanced Software Engineering — Walmart |
| • AI & Azure AI Fundamentals — Microsoft | • Data Analytics Simulations — Accenture & Deloitte |
| • Generative AI Essentials — LinkedIn & Microsoft | • Software Engineering Simulation — Wells Fargo |